Lab 12: Hibernate Associations and Inheritance Strategies

Aim: To study Hibernate Associations and Inheritance Strategies

Programming Exercise:

1. You are given a Hibernate One-to-Many project named *"2.OneToManyEmployee"* using the entities Employee and Department. Import and run the project.
   Then modify the project to implement the following system.
   Entities
   Library
   - libraryId
   - libraryName

   Book
   - bookId
   - bookTitle
   - authorName

   Relationship
   One Library can have many Books.
   Task:
   - Replace Department → Library
   - Replace Employee → Book
   - Update all annotations accordingly.
   - Insert one library with 3–4 books.

   ## Code :

   Book.java

```
package entities;

import jakarta.persistence.CascadeType;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.ManyToOne;
```

```java
import jakarta.persistence.Table;

@Entity
@Table(name = "book_table")
public class Book {
   @Id
   @GeneratedValue(strategy = GenerationType.IDENTITY)
   private int bookId;
   private String bookTitle;
   private String authorName;

   @ManyToOne(cascade = CascadeType.ALL)
//  @ManyToOne(cascade = { CascadeType.MERGE, CascadeType.PERSIST,
CascadeType.REFRESH })
   //@JoinColumn(name = "emp_dept_id")
   private Library book_lib;

   public Book() {
      super();
   }

   public Book(String bookTitle, String authorName) {
      super();
      this.bookTitle=bookTitle;
      this.authorName=authorName;
   }

   public int getBookId() {
      return bookId;
   }

   public String getBookTitle() {
      return bookTitle;
   }

   public String getAuthorName() {
      return authorName;
   }
```

```
   public void setBookTitle(String bookTitle) {
      this.bookTitle = bookTitle;
   }

   public void setAuthorName(String authorName) {
      this.authorName = authorName;
   }

   public void setBook_lib(Library book_lib) {
      this.book_lib = book_lib;
   }

   @Override
   public String toString() {
      return "Book [Id=" + bookId + ", Title=" + bookTitle + ", Author_Name="
+ authorName
            + ", book_lib=" + book_lib.getLibraryName() + "]";
   }
}
```

Library.java

```
package entities;

import java.util.ArrayList;
import java.util.List;

import jakarta.persistence.CascadeType;
import jakarta.persistence.Entity;
import jakarta.persistence.GeneratedValue;
import jakarta.persistence.GenerationType;
import jakarta.persistence.Id;
import jakarta.persistence.OneToMany;
import jakarta.persistence.Table;

@Entity
@Table(name = "library_table")
```

```java
public class Library {
   @Id
   @GeneratedValue(strategy = GenerationType.IDENTITY)
   private int libraryId;
   private String libraryName;
   //@OneToMany(cascade = CascadeType.ALL)
   @OneToMany(mappedBy = "book_lib", cascade = CascadeType.ALL)
//  @OneToMany(cascade = CascadeType.ALL)
   private List<Book> books;

   public Library() {
      super();
      this.books = new ArrayList<>();
   }

   public Library(String libraryName) {
      super();
      this.libraryName = libraryName;
      this.books = new ArrayList<>();
   }

   public int getLibraryId() {
      return libraryId;
   }

   public String getLibraryName() {
      return libraryName;
   }

   public void setLibraryName(String libraryName) {
      this.libraryName = libraryName;
   }

   public List<Book> getBooks() {
      return books;
   }

   public void setBooks(List<Book> books) {
```

```
        this.books = books;
    }

    @Override
    public String toString() {
        return "Library [libId=" + libraryId + ", lib_name=" + libraryName + ",
books=" + books
                + "]";
    }
}
```

BookDAO.java

```
package repository;

import org.hibernate.Session;

import entities.Book;
import utilities.HibernateUtil;

public class BookDAO {
    private final Session session;

    public BookDAO() {
        super();
        session = HibernateUtil.getSessionFactory().openSession();
    }

    public void addBook(Book emp) {
        session.getTransaction().begin();
        session.persist(emp);
        session.getTransaction().commit();
    }

    public void showBook(int Book_id) {
        Book temp_book = session.get(Book.class, Book_id);
        if (temp_book != null)
            System.out.println(temp_book);
```

```
      else
         System.out.println("Book is not valid");
   }

   public void updateBookTitle(int Book_id, String Book_title) {
      Book temp_book = session.get(Book.class, Book_id);
      if (temp_book != null) {
         temp_book.setBookTitle(Book_title);
         session.getTransaction().begin();
         session.merge(temp_book);
         session.getTransaction().commit();
      }
   }

   public void deleteBook(int Book_id) {
      Book temp_book = session.get(Book.class, Book_id);
      if (temp_book != null) {
         session.getTransaction().begin();
         session.remove(temp_book);
         session.getTransaction().commit();
      }
   }
}
```

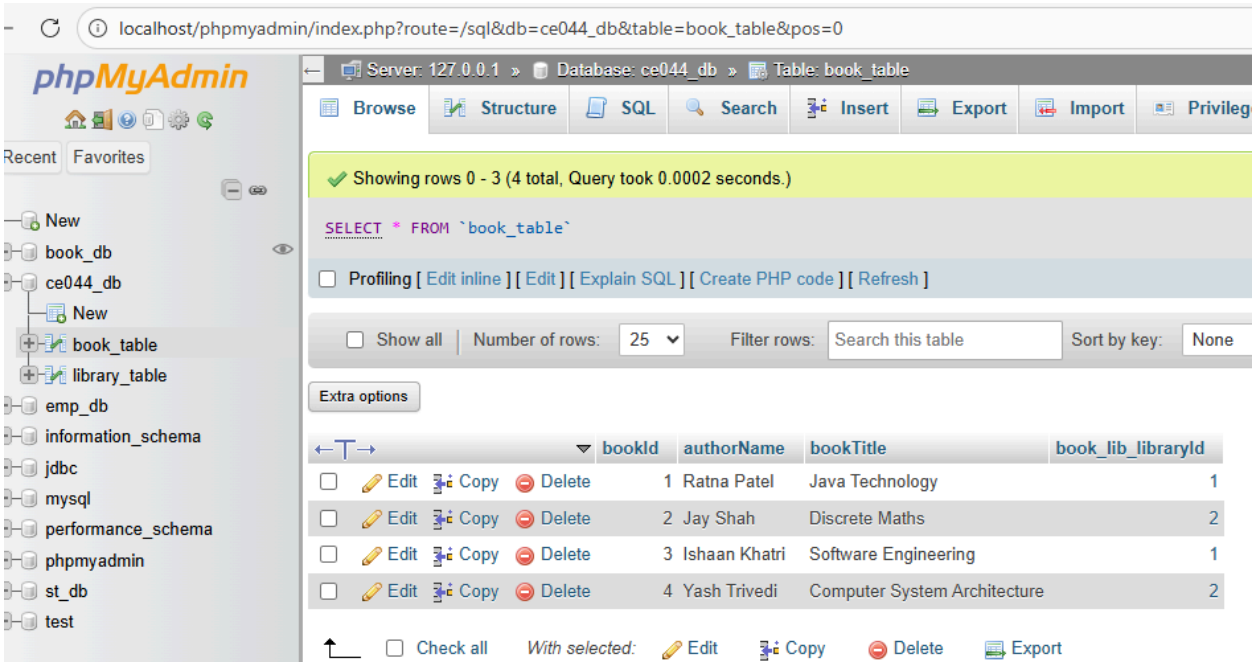
LibraryDAO.java

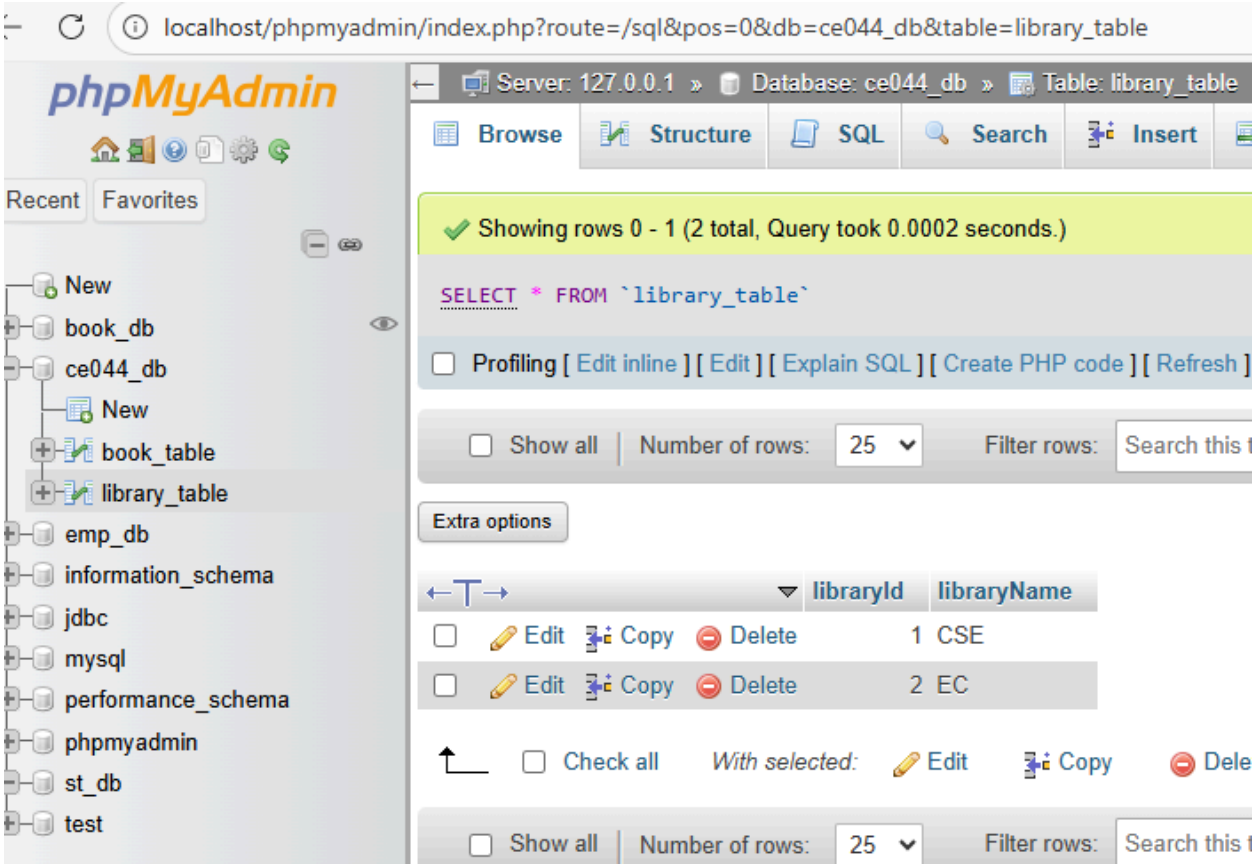
```
package repository;

import java.util.List;

import org.hibernate.Session;

import entities.Library;
import entities.Book;
import utilities.HibernateUtil;

public class LibraryDAO {
   private final Session session;
```

```java
public LibraryDAO() {
    super();
    session = HibernateUtil.getSessionFactory().openSession();
}

public void addLibrary(Library dept) {
    session.getTransaction().begin();
    // you may check if there is already a Library with same name
    session.persist(dept);
    session.getTransaction().commit();
}

public void addBookToBooks(int libraryId, Book book) {
    Library temp_lib = session.get(Library.class, libraryId);
    if (temp_lib != null) {
        book.setBook_lib(temp_lib);

        List<Book> staff_list = temp_lib.getBooks();
        staff_list.add(book);

        session.getTransaction().begin();
        session.persist(book);
        session.getTransaction().commit();
    }
}

public void showLibrary(int libraryId) {
    Library temp_lib = session.get(Library.class, libraryId);
    if (temp_lib != null)
        System.out.println(temp_lib);
}

public void updateLibrary(int libraryId, String libName) {
    Library temp_lib = session.get(Library.class, libraryId);
    if (temp_lib != null) {
        temp_lib.setLibraryName(libName);
        session.getTransaction().begin();
```

```
            session.merge(temp_lib);
            session.getTransaction().commit();
        }
    }

    public void deleteLibrary(int libraryId) {
        Library temp_lib = session.get(Library.class, libraryId);
        if (temp_lib != null) {
            session.getTransaction().begin();
            session.remove(temp_lib);
            session.getTransaction().commit();
        }
    }
}
```

Output :





2. You are given a Hibernate Inheritance strategy project named *"inheritance_singletable_strategy"* using the entities Vehicle, Car and Bike. Import and run the project.
   Then modify the project to implement the following system.

   Use the following classes:
   - Account (Parent Class) in place of Vehicle
   - SavingAccount (Child Class) in place of Car
   - CurrentAccount (Child Class) in place of Bike

   Attributes of tables:

   Account
   - accountId
   - holderName

   SavingAccount
   - interestRate

   CurrentAccount
   - overdraftLimit

Insert 2 records for each account type.

Code :

Account.java

```
package Entities;

import jakarta.persistence.*;

@Entity
@DiscriminatorValue("BIKE")
public class CurrentAccount extends Account {

   private int overdraftLimit;

   public int getOverdraftLimit() {
       return overdraftLimit;
   }

   public void setOverdraftLimit(int overdraftLimit) {
       this.overdraftLimit = overdraftLimit;
   }
}
```

SavingAccount.java

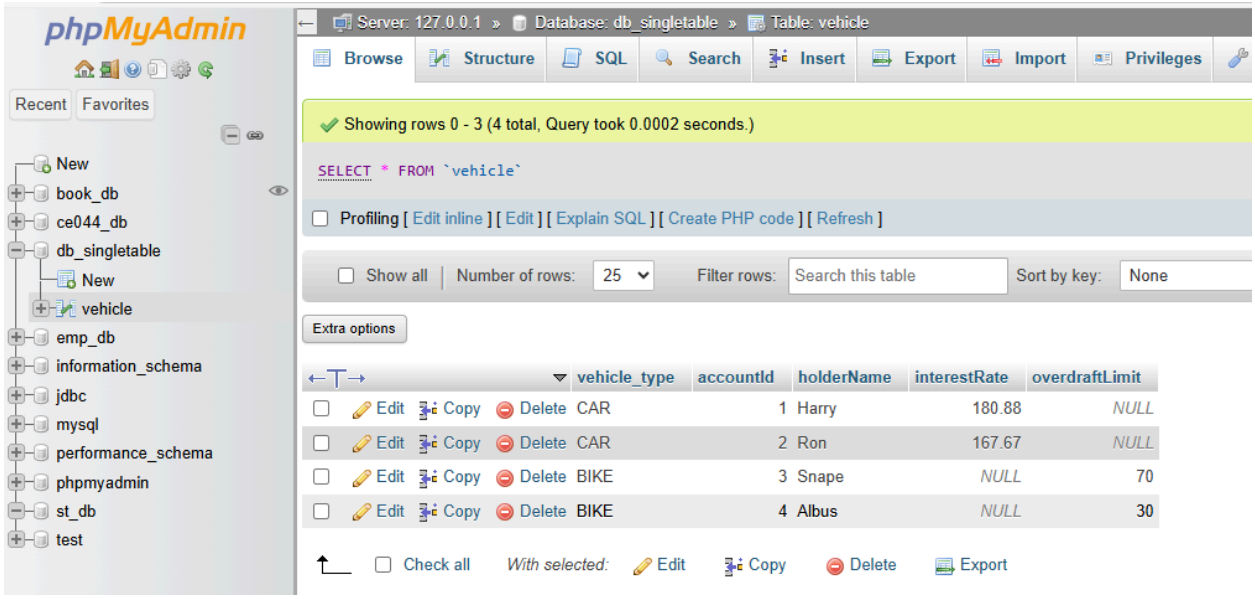
```
package Entities;

import jakarta.persistence.*;

@Entity
@DiscriminatorValue("CAR")
public class SavingAccount extends Account {

   private double interestRate;

   public double getInterestRate() {
       return interestRate;
   }

   public void setInterestRate(double interestRate) {
       this.interestRate = interestRate;
   }
}
```

CurrentAccount.java

```java
package Entities;

import jakarta.persistence.*;

@Entity
@DiscriminatorValue("BIKE")
public class CurrentAccount extends Account {

   private int overdraftLimit;

   public int getOverdraftLimit() {
       return overdraftLimit;
   }

   public void setOverdraftLimit(int overdraftLimit) {
       this.overdraftLimit = overdraftLimit;
   }
}
```

Main.java

```java
package Driver;
import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.cfg.Configuration;

import Entities.Account;
import Entities.SavingAccount;
import Entities.CurrentAccount;

public class Main {

   public static void main(String[] args) {

       SessionFactory factory = new Configuration()
               .configure()
               .addAnnotatedClass(Account.class)
               .addAnnotatedClass(SavingAccount.class)
               .addAnnotatedClass(CurrentAccount.class)
               .buildSessionFactory();

       Session session = factory.openSession();
       session.beginTransaction();

       SavingAccount savingAccount1 = new SavingAccount();
       savingAccount1.setHolderName("Harry");
       savingAccount1.setInterestRate(180.88);
```

```
        SavingAccount savingAccount2 = new SavingAccount();
        savingAccount2.setHolderName("Ron");
        savingAccount2.setInterestRate(167.67);

        CurrentAccount currentAccount1 = new CurrentAccount();
        currentAccount1.setHolderName("Snape");
        currentAccount1.setOverdraftLimit(70);

        CurrentAccount currentAccount2 = new CurrentAccount();
        currentAccount2.setHolderName("Albus");
        currentAccount2.setOverdraftLimit(30);

        session.save(savingAccount1);
        session.save(savingAccount2);

        session.save(currentAccount1);
        session.save(currentAccount2);

        session.getTransaction().commit();
        session.close();
        factory.close();

        System.out.println("Records inserted successfully");
    }
}
```

Output :